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NEW PALEOMAGNETIC DATA FROM CARBONIFEROUS VOLCANICS AND RED BEDS FROM CENTRAL NEW BRUNSWICK

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Abstract. It has long been argued, on the basis of paleomagnetic data derived from Devonian, Carboniferous and Permian lithological units from eastern North America that a sinistral megashear of Carboniferous age parallels and lies within the faults delimiting the Appalachian-Caledonian chains. Recent paleomagnetic studies in the northern Appalachians have cast doubt on the models of a Carboniferous offset. The geologic evidence suggests that only a small amount of dextral (and not sinistral) strike slip has occurred on faults which are parallel with pre-Acadian paleogeographic realms. The purpose of this paleomagnetic study is to test the validity of the proposed left-lateral motion at one of the sites of its presumed passage in central New Brunswick. Six sites (80 oriented specimens) in Carboniferous red beds and volcanics were collected on both sides of the Fredericton Fault and other parallel faults. After AF and thermal cleaning, the mean direction of magnetization is 158°, +38°; the fold test is indecisive. The corresponding paleopole is 135°E, 21°N and the paleolatitude 20°S. The paleopole positions and paleolatitudes are not significantly different on either side of the Fredericton fault and no left-lateral motion was detected by paleomagnetic means. As no such motion was detected in south-central New Brunswick and Newfoundland, it is possible but unlikely that it took place in northern New Brunswick and in the western Gaspé Peninsula. This contribution is useful in discussing possible motions of Acadia.

Introduction

On the basis of paleomagnetic data from eastern North America, it has been argued that a Carboniferous sinistral megashear is present within and parallel to the northern Appalachian chain (Kent and Opdyke, 1979). This megashear is presumably located in Maine and New Brunswick (Sparlosu and Kent, 1983).
The presence or absence of this megashear is of real importance since it is bound to change the paleogeographic restoration of the northern Appalachians versus the North American craton in Carboniferous time (Kent and Opdyke, 1979; Bradley, 1983). In New Brunswick, only three faults are eventual candidates for this megashear: the Catamaran, Norumbega-Fredericton and Beleisle faults. The Norumbega-Fredericton (NF) fault is most acceptable for this eventual megashear in view of its widespread tectonic break and because it transects Carboniferous lithological units over the whole length of the Carboniferous cover (Fig. 1).

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The objective of this study is to test the hypothesis of a megashear coinciding with the NF fault with the aid of a paleomagnetic survey carried out on correlative Carboniferous lithological units located on both sides of the fault (Seguin, 1982; Seguin and Fyffe, 1983). In order to do so, paleolatitudes have been calculated and compared with each other.

Geological Setting and Sampling Sites

The Carboniferous rocks of New Brunswick occupy a successive basin within the Appalachian Orogen (Poole, 1976). The flat-lying sequence consists of continental sedimentary and minor volcanic rocks. The localities sampled are situated in Boiestown, Royal Road quarry and Minto (Fig. 1). Two first localities (4 sites) are situated on the northwestern side of the NF fault and a last locality (2 sites) to the southeast. The age of samples on the NW side is the same as for those on the SE side: upper Namurian (B-C)-lower Westphalian (A) (Van de Poll, 1973). The age control is based on paleontological data obtained from under-, overlying sedimentary sequences. The basalts display subhorizontal layering; magnetite is abundant and hematite present in small amounts.

NRM Directions and Intensities

NRM directions for volcanics are most frequently oriented in the SE quadrant and a few directions in the volcanics coincide with the present

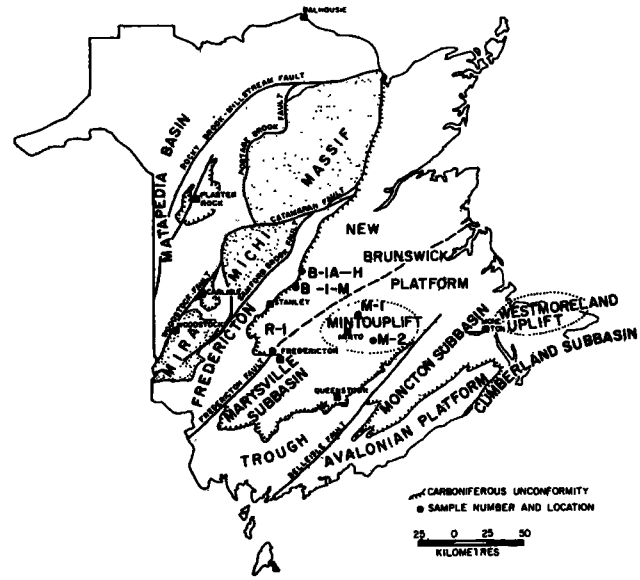


Fig. 1. Tectonostratigraphic zones of New Brunswick and distribution of paleomagnetic sampling localities. Site R2 is close by R1.

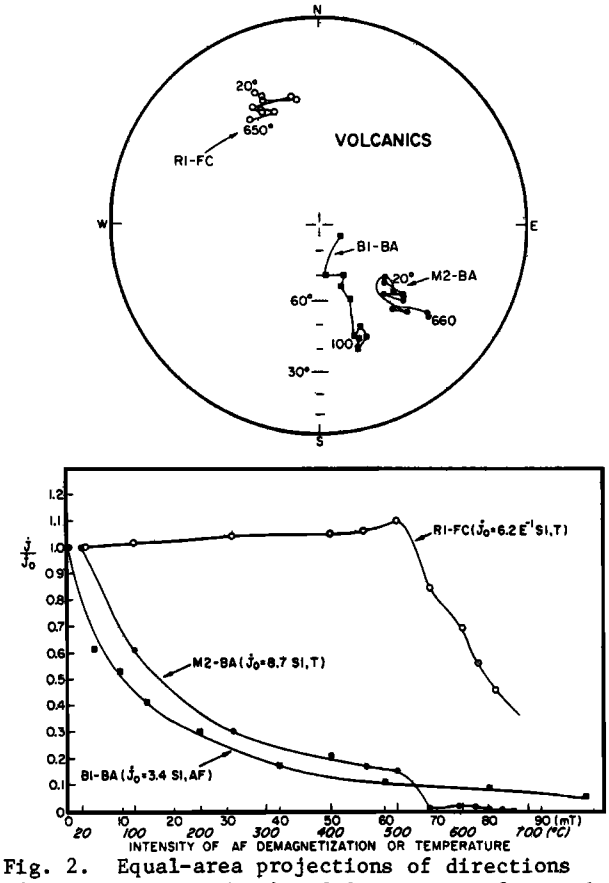


Fig. 2. Equal-area projections of directions of magnetization (top) of 3 specimens from volcanic samples after stepwise AF and thermal treatments. The numbers next to the points represent the demagnetization steps in mT or °C as indicated. The respective normalized AF and thermal decay curves are shown on the bottom.

Earth's field (PEF). An old component of magnetization was preserved in the volcanics since an antiparallel direction is occasionally present in the NW quadrant. Koenigsberger's ratios (Q_n) are close to unity.

Demagnetizations

All Carboniferous volcanics and red beds were demagnetized thermally (20 - 660°C) and with alternating field (AF) (0-100mT). Fig. 2 shows variations in orientation and intensity of magnetization for samples originating from three different sites of volcanics (B1, R1 and M2). Sites B1 and R1 are located on the same side of the NF fault while sites B1 and M2 on opposite side have a similar direction of magnetization (SSE, shallow to intermediate downward inclination). Site R1 has an opposite direction (NNW, shallow to intermediate upward inclination). The SSE directions are characterized by a median destructive field ($H_{1/2}$) of 10 - 20 mT and unblocking temperature (T_{UB}) of 450 - 600°C. Specimen B1-BA (Fig. 2) shows an example of the 11 specimens containing a low coercivity PEF component. For the NNW directions, $H_{1/2}$ values range from 20 to 40 mT and T_{UB} values between 500 and 640°C, that is slightly higher coercivities and T_{UB}s for the NNW directions of magnetization.

A SSE cleaned direction of magnetization similar to that of the volcanics is obtained upon AF thermal demagnetizations for the red beds. Sites B-1-(A) and R-2 (close by R-1) are located on the northern side of the NF fault and site M-1 to the south. Fig. 3 shows an example of variation in direction and intensity of magnetization drawn from samples on sites B1 and M1. Only one component of magnetization is present. The memory carrier is located in both magnetite and hematite which recorded the same component.

Mean Direction of Magnetization

Fig. 4 shows the sample-mean directions for both volcanics and red beds. Two groups are observed: one of normal polarity in the NNW quadrant and another of reverse polarity in the SSE quadrant (Table 1). The SSE directions of the volcanics (153°, +37°) is not significantly different from those of the red beds (164°, +38°). The largest difference in mean inclination observed between sites B1 (B1-A) and R-1 (R-2) (volcanics or red beds, in-situ attitude) is 11° (Table 1). As the α_{95} values vary between 6° and 12°, it is clear that B and R sites witnessed the same event of remanence acquisition. Clearly, any tectonic offset smaller than the maximum dispersion between the sites (≈1000 Km) remains paleomagnetically undetected. The presence of a univectorial component, the coincidence of the SSE directions for the volcanics and red beds and

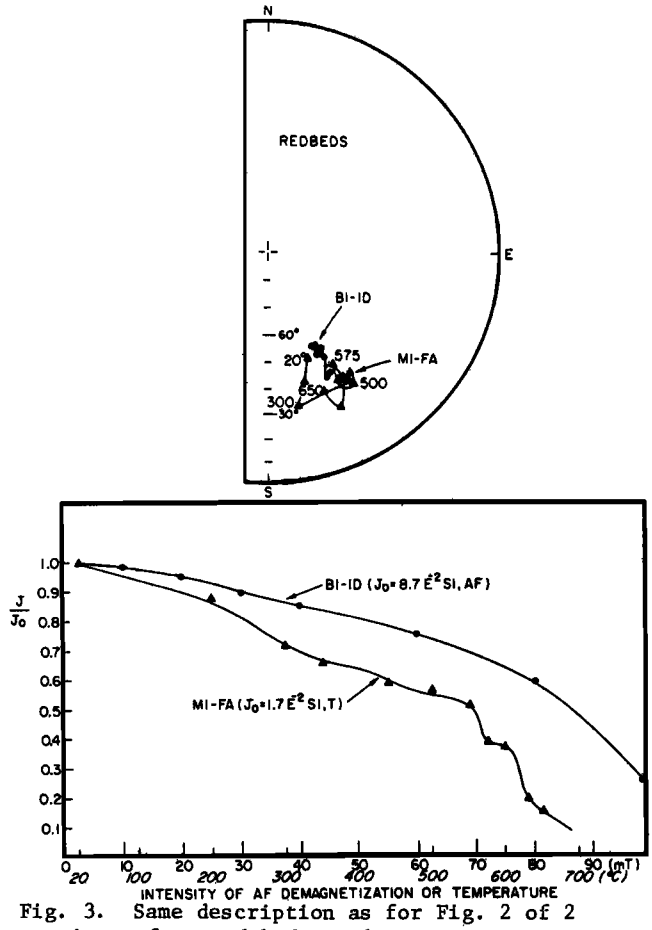


Fig. 3. Same description as for Fig. 2 of 2 specimens from red bed samples.

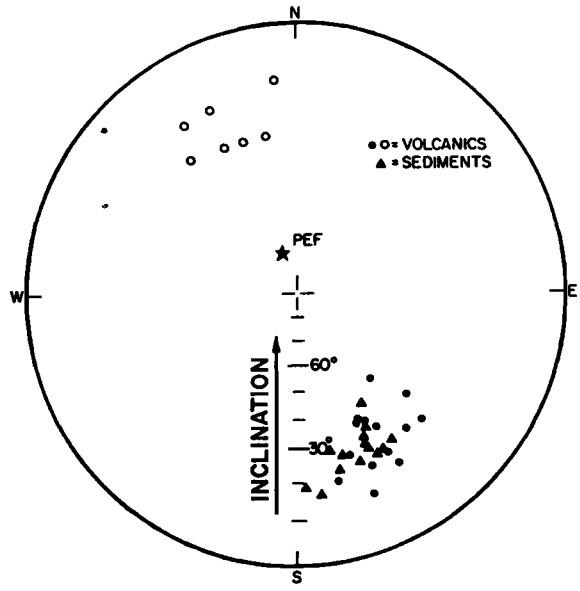
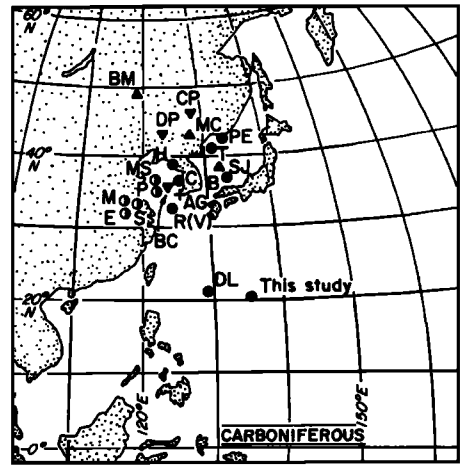


Fig. 4. Equal-area projections of sample-mean directions of magnetization obtained for volcanics and red beds after AF and thermal treatment but without tilt correction (see Table 1).



PALEOPOLES DERIVED FROM CARBONIFEROUS ROCK UNITS. RE-DRAWN FROM ROY ET AL. 1983; ADDITION: DL (IRVING AND STRONG, 1984)

Fig. 5. Paleopoles derived from Carboniferous rock units. Redrawn from Roy et al. (1983). Addition: DL (Irving and Strong, 1984).

the reversals observed in the volcanics suggest a primary component of magnetization.

Fold Test

The strikes and dips of the formations and the results of the in-situ and tilt corrected mean directions of magnetization are shown on Table 1. The degree of dispersion is slightly increased after tilt correction but the fold test is not decisive. In fact, the tilt correction may not be pertinent to this study; there are structural indications that flow layering in volcanics was not originally horizontal but made an angle of 5° to 10° with respect to the horizontal in the vicinity of volcanic centres. In the red beds, it is difficult to decipher the exact horizontal plane. For these reasons, the in-situ component of magnetization is thought to be more meaningful than the tilt- corrected one.

Comparison of the Middle Carboniferous New Brunswick Pole with Respect to Cratonic North American Poles.

The paleopole obtained for the volcanics and red beds is 135°E, 21°N; the paleolatitude is 20° S. There is no significant difference in paleopole and paleolatitude between blocks situated on either side of the NF fault. The suggestion that Acadia was displaced northwards by about 1500 Km relative to cratonic North America in the Carboniferous along or in the vicinity of the NF fault is not supported by the present paleomagnetic data. The demonstration of little or no post-Namurian B-C motion along the NF does not dismiss the possibility that the motion between Acadia and

TABLE 1. Thermally and AF Cleaned Directions of Magnetization

SITE NO	N	n	STRIKE (°)	DIP (°)	IN-SITU D (°)	I (°)	α_{95} (°)	K	TILT-CORRECTED D (°)	I (°)	α_{95} (°)	K
						VOLCANICS						
B-1	7	15	040	5SE	153	44	9.7	39	151	39		
R-1*	7	16	050	15SE	158	32	12.1	33	156	19		
M-2	7	18	060	10SE	149	35	8.3	53	149	25		
MEAN:	3				153	37	11.0	127	152	28	16.5	60
						RED BEDS						
B-1A	5	17	005	4W	153	44	6.9	123	156	46		
R-2	2	5	050	15SE	169	33	---	---	165	21		
M-1	7	16	045	5SE	163	37	6.2	93	161	33		
MEAN:	3				164	38	13.5	82	161	33	20.0	40
MEAN:	6	(VOLCANICS - RED BEDS)			158	38	6.5	104	156	31	9.7	49

* NORMAL POLARITY

North America could have occurred in early Carboniferous time or that it could have taken place in late Carboniferous in other regions of the Canadian Maritimes. On geological ground, the faults observed in the Fredericton Trough suggest a small right lateral rather than left lateral motion. The limitations of the paleomagnetic method and the small quantity of paleomagnetic studies are such that the possibility of small displacements (hundreds of kms) cannot be discounted.

Recent paleomagnetic data obtained from early Carboniferous rocks from cratonic North America (Deer Lake Group) in western Newfoundland (Irving and Strong, 1984) and from the Avalon platform in eastern Newfoundland (Terrenceville Formation; Kent, 1982) yielded paleopoles and paleolatitudes (15-20°S) statistically identical to the results presented in this study (Fig. 5).

The inclinations of the present paleomagnetic study agree with those in older Carboniferous (Viséan to possibly early Namurian (A)) rocks from Newfoundland the inclinations of which are about 15° higher than those in southern New Brunswick and Nova Scotia (e.g. Roy and Morris, 1983) which are mainly late Namurian (C) to early Westphalian (A). The paleolatitudes (8-10°S) from the southern New Brunswick - Nova Scotia red bed basin are lower than those from Carboniferous rocks of central New Brunswick and Newfoundland. The divergence can be explained by a systematic variation of paleolatitude (i.e. a sudden apparent polar wander, APW) of the Appalachians and the North American craton with a motion towards the north during middle Carboniferous time or by tectonic motion of southern New Brunswick and Nova Scotia with respect to central and northern New Brunswick as well as Newfoundland.

Which one of these two scenarios is more plausible? What are the arguments for or against? If the displacement of Acadia took place in late Devonian or early Carboniferous time, then there is no reason why middle and late Carboniferous red beds from southern New Brunswick and Nova Scotia (Acadia) should have recorded a 10° to 13° southern paleolatitude difference with respect to central and northern New Brunswick and Newfoundland. There is always a remote possibility that displacement of Acadia took place by Middle to Late Carboniferous time but geological observations do not concord with this hypothesis. For these reasons, it is more probable but still incompletely proven that a rapid APW took place at late Namurian (C) to early Westphalian (A) time.

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